

Paul Livingston: Tractatus

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Wittgenstein's *Tractatus Logico-Philosophicus* draws a limit to language by examining the possibilities and impossibilities of thought using his picture theory to find the limits of reality. He does this to show not just the limits of language but the limits of our world. To find limits to things is a responsibility of philosophy, specifically "[p]hilosophy should make clear and delimit sharply the thoughts which otherwise are, as it were, opaque and blurred."

Wittgenstein's famously written *Tractatus Logico-Philosophicus* argues thought, language, and the world are isomorphic. They are isomorphic because they each share the same logical form, which is $(\bar{p}, \bar{\xi}, N(\bar{\xi}))$. The isomorphism of the elements thought, language, and the world in unison accurately represent reality. If something falls outside of this unison, it is nonsense. Therefore, the *Tractatus Logico-Philosophicus* draws a limit to language by examining the possibilities and impossibilities of thought to find the limits of reality.

First, an isomorphism can be expressed as a mathematical function which represents a one to one mapping between things of the same type. This means each element from one set can be paired with a single element from another set. For instance, if I have the set $\{1, 2, 3, 4, 5\}$ and I take $f(x) = x + 1$, then my isomorphic set is $\{2, 3, 4, 5, 6\}$. Each element in the first set can directly correspond to an element in the second set. An isomorphism essentially maintains the relationship between elements of the sets.

For Wittgenstein, the world, thought, and language are isomorphic. Therefore, drawing a limit to any one of these three draws a limit to all three. Wittgenstein connects logic to

the world by stating that the “specification of all true elementary propositions describes the world completely” (Tractatus 4.25). He connects reality to thought by calling a thought a “picture” — “The logical picture of facts is the thought” (Tractatus 3) and “The totality of true thoughts is a picture of the world” (Tractatus 3.01) connect thought to the world. Finally - he connects language with logic by saying the “totality of propositions is the language” (Tractatus 4.001). To further drive this point home, and segue into the reason that Wittgenstein would draw limits, we have 5.6: “The limits of my language mean the limits of my world.”

If the totality of all true elementary propositions describes the world complete, then one needs only to analyze these propositions to understand the world in its entirety. As Wittgenstein states, there are *exponentially* more possible states than there are elementary propositions (for n logical propositions there are 2^n (or $\sum_{v=0}^n \binom{n}{v}$) possibilities (Tractatus 4.27). This is an absurd amount and is clearly not possible to analyze directly. We therefore must make generalizations that apply *regardless* of the state or number of possibilities. One such generalization is that we are able to use propositional logic to describe everything about the world. Therefore, any limitation of propositional logic also applies to the world, or at least to our capability to describe and understand the world.

Where, then, do these limits lie? Wittgenstein uses the example of the eye and the field of view (Tractatus 5.6331). The eye can see everything within its field of view, but not the eye itself. Likewise, a picture as described in Wittgenstein’s *Tractatus* cannot describe

itself, as it then becomes a picture of a picture — a higher-order picture, and then is no longer describing itself. If we had a picture of the world, we would *still* not have a picture of this picture. This problem extends to all realms that we can conceive of. Even if there is an afterlife, this problem would persist in it if it resembles our current state!

There are certainly valid objections to this. Certain *recursively enumerable* languages and complete systems which are capable of describing themselves show that one does not need to be “outside” a picture as Wittgenstein suggests to describe it fully. For example, the C programming language is itself written in C. It stands to reason that any Turing-complete language should be capable of fully describing itself. Therefore, this idea that one must be “outside” a picture to fully describe it is not quite true. Some systems may be fully described from within. A definition of our world may be much more like this than like an eye.

Another objection to Wittgenstein’s way of describing the world is the idea that first-order logic is not sufficient to fully describe language, or all truths about the world for that matter. Wittgenstein touches on paradoxes (he shows his solution to Russell’s Paradox in 3.333) but considers anything “unlogical” not to be expressible. Perhaps paradoxes are not “unlogical”, or perhaps they are unnecessary. Furthermore, we know thanks to Gödel that there are truths about basic arithmetic which cannot be proven by a consistent first-order logic. This opens up the possibility that the methodology described in the *Tractatus* is really only capable of partially describing the world.

Nonetheless, if we accept this view that first-order logic is capable of describing the world, we certainly find the world to have limits. Anything outside these limits is “nonsense,” and not only that, but it is quite impossible to speak of or even conceive of these things. While we can be shown the limits, we cannot go beyond them as they are not of this world. Why have these limits? Wittgenstein provides an answer in 5.1362 — “The freedom of the will.” Causality is one such thing that is outside the limits of our world, while we might accept causality casually as self-evident, evidence does not prove the existence of causality. If there is causality, then we run the risk of having a fully deterministic world, one devoid of free will in which we are just passengers. Wittgenstein was motivated to dodge this interpretation of our universe.

While the capability to doubt causality is not novel in the *Tractatus*, there are certainly some new and interesting insights gleaned from this. For example, in 6.4311 Wittgenstein says that “he lives eternally who lives in the present.” As there is no causality, and we have no concept of the passage of time other than our own perception and the instruments we use to measure it, he states we can consider eternity to be a “timelessness” rather than an endless temporal duration.

We can extend this doubt to the entire field of metaphysics — it is all beyond the limit, or “nonsense.” The *Tractatus* ultimately acts as a tool for us to solve the problems of metaphysics. We solve the problems of metaphysics not by solving them, but by showing them to be unsolvable, a mere temptation to pursue nonsense.